
Natural Language Processing

Lecture-3: Deep Learning methods for NLP

❖ SUMMARY

Framework

We aim to model the sequence $(X_1, \dots, X_T)$ in order to predict the label Y .

$$\begin{aligned} dnn_{\theta}: \mathbb{R}^{d \times T} &\rightarrow \mathbb{R}^p \text{ or } [[0, K]]^p \\ (X_1, \dots, X_T) &\mapsto \hat{Y} \end{aligned}$$

- The output space is $\mathbb{R}^p$ for regression tasks.
- The output space is $[[0, K]]^p$ for classification tasks.
- Given a sequence $(X_1, \dots, X_T)$, the objective is to predict Y .

Most deep learning models:

- Are parametric, with parameters $\theta \in \mathbb{R}^D$.
- Are defined as compositions of simple functions (linear and non-linear).
- Are trained end-to-end using backpropagation.

Note: In deep learning, the parametrization of dnn_{θ} is referred to as the *architecture*.

Types of Architectures

The predictive function dnn_{θ} can be defined using different architectures:

- Multi-Layer Perceptron (MLP)
- Recurrent Neural Networks (RNNs)
- Attention-based models
- Self-Attention models (Transformers)

Training Method:

- Stochastic Gradient Descent (SGD), commonly implemented via backpropagation.

Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron is the simplest deep learning architecture. It operates on fixed-size (unidimensional) input data of dimension d .

Regression Setting:

$$dnn_{\theta}: \mathbb{R}^d \rightarrow \mathbb{R}^2, X \mapsto \hat{Y}$$

Model Definition:

$$dnn_{\theta}(X) = W_2 \varphi_1(W_1 X + b_1) + b_2$$

Parameters and Dimensions:

- $W_1 \in \mathbb{R}^{\delta \times d}$
- $b_1 \in \mathbb{R}^{\delta}$
- $W_2 \in \mathbb{R}^{2 \times \delta}$
- $b_2 \in \mathbb{R}$
- $\varphi_1: \mathbb{R}^{\delta} \rightarrow \mathbb{R}^{\delta}$ is a fixed non-linear activation function

Key Characteristics:

- A 2-layer neural network
- Contains one hidden layer of size δ
- Maps an input of dimension d to an output of dimension 2
- Also known as a *Feed-Forward Neural Network (FNN)*

❖ MCQ

1. What does the model aim to learn when modeling $(X_1, \dots, X_T)$?
A. Only the input sequence
B. The correct label Y
C. The hidden layers only
D. The parameters only

Answer: B

2. The function dnn_{θ} maps from:
A. $\mathbb{R}^T \rightarrow \mathbb{R}^d$
B. $\mathbb{R}^{(d,T)} \rightarrow \mathbb{R}^p$ or $[[0,K]]^p$
C. $\mathbb{R}^p \rightarrow \mathbb{R}^{(d,T)}$
D. $\mathbb{R}^d \rightarrow \mathbb{R}^T$

Answer: B

3. The output $\hat{Y}$ represents:
A. The true label
B. The predicted label
C. The hidden representation
D. The loss value

Answer: B

4. For regression tasks, the output space is:
A. $[[0,K]]^p$
B. $\mathbb{R}^d$
C. $\mathbb{R}^p$
D. $\mathbb{R}^T$

Answer: C

5. For classification tasks, the output space is:
A. $\mathbb{R}^p$
B. $[[0,K]]^p$
C. $\mathbb{R}^{(d,T)}$
D. $\mathbb{R}^d$

Answer: B

6. Deep Learning models are generally:
- A. Non-parametric
 - B. Parametric with $\theta \in \mathbb{R}^D$
 - C. Only linear
 - D. Only probabilistic
- Answer: B
7. Deep Learning models are defined as:
- A. A single linear function
 - B. A composition of simple linear and non-linear functions
 - C. Only non-linear transformations
 - D. Random mappings
- Answer: B
8. How are most Deep Learning models trained?
- A. Rule-based systems
 - B. Manual tuning
 - C. End-to-end with backpropagation
 - D. Random initialization only
- Answer: C
9. In Deep Learning, the parametrization of dnn is called:
- A. Optimization
 - B. Architecture
 - C. Dataset
 - D. Activation
- Answer: B
10. Which of the following is NOT a type of architecture mentioned?
- A. Multi-Layer Perceptron
 - B. Recurrent Layers
 - C. Convolutional Layers
 - D. Attention Layers
- Answer: C
11. Self-attention layers are used in:
- A. MLP
 - B. CNN
 - C. Transformer Architecture
 - D. RNN only
- Answer: C
12. How are model parameters estimated?
- A. Linear regression
 - B. Stochastic Gradient Descent (backpropagation)
 - C. Clustering
 - D. Sampling
- Answer: B
13. The Multi-Layer Perceptron (MLP) is considered:
- A. The most complex architecture
 - B. The simplest Deep Learning architecture
 - C. A probabilistic model
 - D. A non-parametric model
- Answer: B
14. The MLP works on:
- A. Sequential data only
 - B. Multidimensional tensors only
 - C. Unidimensional data

D. Graph data

Answer: C

15. In the given example, the MLP is presented for:

A. Classification

B. Regression

C. Clustering

D. Reinforcement learning

Answer: B

16. The function of MLP in this case is:

A. $\mathbb{R}^2 \rightarrow \mathbb{R}^d$

B. $\mathbb{R}^d \rightarrow \mathbb{R}^2$

C. $\mathbb{R}^T \rightarrow \mathbb{R}^p$

D. $\mathbb{R}^p \rightarrow \mathbb{R}^T$

Answer: B

17. The MLP equation is:

A. $W_1X + b_1$

B. $W_2X + b_2$

C. $W_2 \phi_1(W_1X + b_1) + b_2$

D. $\phi_1(W_2X + b_2)$

Answer: C

18. Which of the following are trainable parameters?

A. ϕ_1 only

B. W_1, b_1, W_2, b_2

C. X only

D. Y only

Answer: B

19. The dimension of W_1 is:

A. $\mathbb{R}^{(d \times \delta)}$

B. $\mathbb{R}^{(\delta \times d)}$

C. $\mathbb{R}^{(2 \times d)}$

D. $\mathbb{R}^\delta$

Answer: B

20. The dimension of b_1 is:

A. $\mathbb{R}^d$

B. $\mathbb{R}^2$

C. $\mathbb{R}^\delta$

D. $\mathbb{R}^p$

Answer: C

21. The dimension of W_2 is:

A. $\mathbb{R}^{(\delta \times 2)}$

B. $\mathbb{R}^{(2 \times \delta)}$

C. $\mathbb{R}^{(d \times \delta)}$

D. $\mathbb{R}^2$

Answer: B

22. The bias b_2 belongs to:

A. $\mathbb{R}^d$

B. $\mathbb{R}^\delta$

C. $\mathbb{R}$

D. $\mathbb{R}^2$

Answer: C

23. The function ϕ_1 is:

A. Trainable linear function

- B. Fixed non-linear function
- C. Random function
- D. Loss function

Answer: B

24. The mapping of ϕ_1 is:

- A. $\mathbb{R}^d \rightarrow \mathbb{R}^2$
- B. $\mathbb{R}^2 \rightarrow \mathbb{R}^\delta$
- C. $\mathbb{R}^d \rightarrow \mathbb{R}^\delta$
- D. $\mathbb{R}^\delta \rightarrow \mathbb{R}^d$

Answer: C

25. The described MLP has:

- A. 1 layer
- B. 2 layers with no hidden layer
- C. 2 layers with 1 hidden layer
- D. 3 hidden layers

Answer: C

26. The hidden layer has dimension:

- A. d
- B. 2
- C. δ
- D. p

Answer: C

27. The model takes input of dimension:

- A. 2
- B. δ
- C. d
- D. p

Answer: C

28. The model outputs a vector of dimension:

- A. d
- B. δ
- C. 2
- D. T

Answer: C

The MLP is also called: .29

- A. CNN
- B. RNN
- C. Feed-Forward Neural Network (FNN)
- D. Transformer

Answer: C